

Executive Business Report

Pharmaceutical HCP Segmentation & Commercial Analytics

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Project Type: End-to-End Data Analytics Project

Tools Used: Python, Pandas, NumPy, Plotly, MySQL, Jupyter Notebook, VS Code

1. Executive Summary

The pharmaceutical industry relies heavily on data-driven decision-making to optimize commercial performance, improve physician engagement, and maximize product adoption. This project simulates a real-world pharmaceutical commercial analytics environment by generating and analysing synthetic healthcare professional (HCP), product, prescription, and survey datasets.

Using Python for data generation, cleaning, and exploratory analysis, along with MySQL for business-oriented SQL analysis, the project identifies key commercial trends, physician prescribing behaviour, regional performance, and product effectiveness. The findings provide actionable recommendations that can support pharmaceutical sales teams and business managers in making informed strategic decisions.

2. Business Problem

Pharmaceutical companies market multiple products across various therapy areas while engaging thousands of healthcare professionals with diverse prescribing behaviours. Without systematic analysis, it becomes difficult to identify:

- High-value physicians
- Best-performing products
- High-revenue regions
- Effective commercial strategies
- Physician adoption patterns

This project aims to address these challenges through structured data analysis and business intelligence techniques.

3. Project Objectives

The primary objectives of this project are:

- Generate realistic pharmaceutical commercial datasets.
 - Clean and validate transactional healthcare data.
 - Perform exploratory data analysis to identify trends and patterns.
 - Analyse business performance using SQL.
 - Identify high-value healthcare professionals.
 - Evaluate therapy area and product performance.
 - Generate business recommendations based on analytical findings.
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4. Dataset Overview

The project consists of four interconnected datasets.

Dataset	Description	Records
HCP Master	Healthcare professional information	2,500
Products	Pharmaceutical product details	18
Prescriptions	Sales and prescription transactions	100,000
Survey	Physician behavioural survey	500+

The datasets were synthetically generated to resemble real-world pharmaceutical commercial data while maintaining realistic business relationships.

5. Data Preparation

Before analysis, the datasets underwent a structured cleaning process to ensure quality and consistency.

The data preparation process included:

- Removal of duplicate records
- Handling of missing values
- Data type validation
- Dataset consistency checks
- Quality verification

Following preprocessing, all datasets contained zero missing values and duplicate records were successfully removed, resulting in clean and analysis-ready data.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted using Python to understand overall business trends and physician behaviour.

The analysis included:

- Revenue by Therapy Area
- Revenue by Region
- Product Performance Analysis
- Specialty Distribution
- Adoption Mindset Distribution
- Correlation Heatmap
- Physician Segmentation
- Regional Performance Comparison

These visualizations helped identify patterns in commercial performance and provided a foundation for further SQL-based business analysis.

7. SQL Business Analysis

Business questions were answered using **25 SQL queries**, covering five major analytical categories.

Business KPIs

- Total Revenue
- Total Prescription Volume (TRx)
- Total New Prescriptions (NRx)
- Average Sales Value
- Total Sales Calls

Product Performance

- Revenue by Therapy Area
- Top Revenue-Generating Products
- Highest TRx Product
- Highest NRx Product
- Average Revenue by Therapy Area

HCP Performance

- Top Revenue-Generating HCPs
- Revenue by Medical Specialty
- Average Prescription Volume
- Average Sales Calls by Specialty
- Top HCPs by New Prescriptions

Regional Analysis

- Revenue by Region
- Revenue by State
- Revenue by Hospital Type
- Revenue by Tier
- Top Cities by Revenue

Advanced SQL Analysis

Advanced SQL concepts such as **Window Functions**, **Ranking**, and **Common Table Expressions (CTEs)** were applied to solve more complex business problems, including product ranking, specialty ranking, identifying above-average revenue products, and analysing revenue by physician adoption mindset.

8. Key Findings

The analysis revealed several important business insights:

- Certain therapy areas consistently generated higher commercial revenue.
 - A small number of pharmaceutical products contributed significantly to total sales.
 - High-performing healthcare professionals accounted for a substantial share of prescription revenue.
 - Commercial performance varied across regions, states, and hospital types.
 - Physician adoption mindset influenced product adoption and revenue generation.
 - Advanced SQL techniques provided deeper insights into product rankings and commercial performance.
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9. Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Increase engagement with high-performing healthcare professionals through targeted relationship management.
 - Allocate additional marketing resources to top-performing therapy areas.
 - Strengthen commercial activities in underperforming regions with growth potential.
 - Optimize sales force deployment based on regional and specialty-level insights.
 - Personalize physician engagement strategies using adoption mindset segmentation.
 - Continuously monitor key commercial KPIs to support data-driven decision-making.
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10. Project Outcomes

This project successfully demonstrated an end-to-end analytics workflow, including:

- Synthetic data generation
- Data cleaning and preprocessing
- Exploratory data analysis
- SQL-based business intelligence
- Insight generation
- Business reporting

The project showcases practical applications of Python and SQL in solving real-world pharmaceutical commercial analytics problems.

11. Skills Demonstrated

Throughout the project, the following technical and analytical skills were applied:

- Data Cleaning
 - Exploratory Data Analysis
 - Business Analytics
 - SQL Querying
 - Joins and Aggregations
 - Window Functions
 - Common Table Expressions (CTEs)
 - Data Visualization
 - Healthcare Analytics
 - Business Reporting
 - Insight Generation
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12. Conclusion

The **Pharmaceutical HCP Segmentation & Commercial Analytics** project demonstrates how structured data analysis can transform large volumes of pharmaceutical sales data into meaningful business insights. By integrating Python, SQL, and business reporting techniques, the project provides a comprehensive view of product performance, physician engagement, and regional commercial trends.

The analytical findings and recommendations can support pharmaceutical organizations in improving commercial effectiveness, optimizing resource allocation, and enhancing

physician engagement strategies. This project also highlights the importance of data analytics in enabling evidence-based decision-making within the healthcare and pharmaceutical industry.

Appendix

Tools & Technologies

- Python
- Pandas
- NumPy
- Plotly
- MySQL
- Jupyter Notebook
- VS Code

Project Repository

<https://github.com/sarimhussain72/Pharma-HCP-Segmentation-Commercial-Analytic>
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